Notes: Jiménez, Ongena, Peydró, & Saurina (Econometrica, 2014)

Hazardous Times for Monetary Policy: What Do Twenty-Three Million Bank Loans Say About the Effects of Monetary Policy on Credit Risk-Taking?

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1 Big picture

- **Question.** Does a low monetary policy rate spur banks to take more credit risk?
- **Core channel.** The paper tests the *bank risk-taking channel* of monetary policy: when short-term rates fall, banks–especially banks with little capital at stake–may lend more aggressively to risky borrowers.
- **Data advantage.** The authors use Spain's credit register, including monthly loan applications, granted loans, bank characteristics, firm credit histories, collateral, and future defaults.
- **Main sample.** The loan-application sample covers 2002:02–2008:12 and contains 241,052 applications by firms to banks that are new to the firm. The second-stage sample contains 38,334 granted applications.
- **Identification challenge.** Monetary policy can change borrower quality, borrower demand, bank balance sheets, and total credit supply. The paper must isolate changes in the *composition of credit supply.*
- **Empirical design.** The design uses a two-stage selection model: first whether a loan application is granted, then the loan outcome conditional on approval. Specifications progressively add time-by-firm and time-by-bank fixed effects.
- **Key coefficient.** The core variable is the triple interaction

$$\Delta\text{Overnight Rate}_{t-1} \times I(\text{Firm Risk}_{ti}) \times \log(\text{Bank Capital}_{t-1,b}).$$

  A positive coefficient means that when the overnight rate falls, low-capital banks expand risky lending more than high-capital banks.
- **Main result.** A 1 percentage point decrease in the overnight rate makes low-capital banks grant more applications to risky firms, commit larger credit amounts, require less collateral, and experience higher future default.
- **Macro horse race.** Long-term rates, securitization, current-account deficits, GDP growth, inflation, and U.S. rates do not overturn the overnight-rate result.
- **Takeaway.** Monetary policy affects not only the volume of credit, but also the risk composition of bank credit supply.

2 Data and measurement

2.1 Why Spain and loan applications are useful

- Spain is a bank-dominated economy, so bank lending is central for firm financing.
- Monetary policy is largely external to Spain during the sample because short-term rates are set by the European Central Bank.
- The credit register observes loan applications, not only granted loans. This is crucial because risk-taking can occur at the approval stage.
- Many firms apply to multiple banks in the same month. This allows the authors to compare different banks' treatment of the same firm at the same time.

Interpretation. If the same firm applies to several banks in the same month, differences in approval and loan terms across banks are much closer to supply-side differences than ordinary cross-sectional lending comparisons.

2.2 Loan application and contract data

- The data come from the Credit Register of the Banco de España.
- The application sample starts in 2002 because requests for borrower information were stored from then onward.
- Each request links a potential borrower to a bank. If the bank later grants a loan within the window, the application is classified as successful.
- The sample focuses on applications to banks that are new to the firm, avoiding simple renewals or evergreen lending by incumbent banks.
- The paper matches applications to loan-level outcomes and bank and firm characteristics.

Interpretation. The paper can observe the full lending funnel: application, approval, committed amount, collateral, and later default. This is the main reason the identification is stronger than studies that use only

outstanding or granted loans.

2.3 Main dependent variables

- **Granting of loan application:** equals 1 if the application made in month t to bank b by firm i is successful and the loan is granted in t to $t+3$.
- **Log credit amount:** log committed loan amount granted in t to $t+3$.
- **Future default:** equals 1 if the firm later defaults with the lending bank.
- **Without collateral:** equals 1 if the granted loan is uncollateralized.

Interpretation. The outcomes move from quantity to risk: approval and amount show the supply of credit, while future default and collateral show whether the supplied credit is riskier.

2.4 Main explanatory variables

- **Firm risk:** equals 1 if the firm had any nonperforming loans in the previous four years. Robustness uses one-, two-, three-, and five-year histories.
- **Bank capital:** log bank capital ratio, based on bank equity and retained earnings over total assets.
- **Overnight rate:** annual change in the Euro Overnight Index Average (EONIA), lagged one period.
- **Bank controls:** size, liquidity, profitability, nonperforming-loan ratio, and bank type.
- **Macro controls:** GDP growth, CPI inflation, Spanish long-term rate, U.S. rates, securitization, and current-account deficit.

Interpretation. Firm risk captures ex ante borrower risk. Bank capital captures bank agency problems: banks with less capital have less "skin in the game" and may shift more risk to outsiders.

3 Models and empirical specifications

3.1 Two-stage selection structure

The empirical problem is a panel version of a type-2 Tobit selection model.

- **First stage:** whether a loan application is granted.
- **Second stage:** the outcome of the granted loan, such as credit amount, future default, or collateral.

Interpretation. Selection matters because risk-taking may already occur when banks decide which applications to approve. A study of granted loans only would miss that part of the channel.

3.2 Baseline specification

The parsimonious model is:

$$\begin{aligned} I(\text{Grant}_{tbi}) &= \alpha_t + \alpha_i + \alpha_b + \beta I(\text{Risk}_{ti}) + \delta \Delta ON_{t-1} I(\text{Risk}_{ti}) \\ &\quad + \gamma \Delta ON_{t-1} \log(\text{BankCap}_{t-1,b}) I(\text{Risk}_{ti}) + \text{Controls}_{tbi} + \varepsilon_{tbi}, \\ \log(\text{CreditAmount}_{tbi}) &= \alpha_t + \alpha_i + \alpha_b + \beta' I(\text{Risk}_{ti}) + \delta' \Delta ON_{t-1} I(\text{Risk}_{ti}) \\ &\quad + \gamma' \Delta ON_{t-1} \log(\text{BankCap}_{t-1,b}) I(\text{Risk}_{ti}) + \text{Controls}_{tbi} + \varepsilon'_{tbi}. \end{aligned}$$

Interpretation. The double interaction asks whether low rates increase lending to risky firms in general. The triple interaction asks whether this increase is stronger for low-capital banks, which is the risk-taking channel.

3.3 Saturated benchmark

The more saturated specification uses time-by-firm and time-by-bank fixed effects:

$$\begin{aligned} I(\text{Grant}_{tbi}) &= \alpha_{ti} + \alpha_{tb} + \gamma \Delta ON_{t-1} \log(\text{BankCap}_{t-1,b}) I(\text{Risk}_{ti}) + \text{Controls}_{tbi} + \varepsilon_{tbi}, \\ \log(\text{CreditAmount}_{tbi}) &= \alpha_{ti} + \alpha_{tb} + \gamma' \Delta ON_{t-1} \log(\text{BankCap}_{t-1,b}) I(\text{Risk}_{ti}) + \text{Controls}_{tbi} + \varepsilon'_{tbi}. \end{aligned}$$

Interpretation. Time-by-firm effects compare banks lending to the same firm in the same month. Time-by-bank effects compare the same bank's treatment of different-risk firms in the same month. This nearly saturates away demand and bank-level supply confounds.

3.4 Sign of the key coefficient

- The paper uses the *change* in the overnight rate.
- The important experiment is a *decrease* in the rate.
- A positive γ means that when the rate decreases, a bank with lower capital expands credit to risky firms more than a bank with higher capital.

Interpretation. This is why a positive triple-interaction coefficient is interpreted as evidence that low monetary rates induce more risk-taking by low-capital banks.

3.5 Horse-race specifications

The paper adds parallel triple interactions with:

- Spanish GDP growth and CPI inflation;
- Spanish 10-year government bond rate;
- U.S. federal funds rate and U.S. 10-year rate;
- bank securitization over assets;
- current-account deficit over GDP;
- bank size, liquidity, profitability, nonperforming-loan ratio, and type.

Interpretation. The goal is to show that the results are not merely driven by long-term rates, global liquidity, securitization, capital inflows, or changing macro conditions.

4 Identification and empirical design

4.1 What identifies the main effect

- **Firm demand and quality:** time-by-firm fixed effects absorb all time-varying firm demand and borrower-quality shocks.
- **Bank supply volume:** time-by-bank fixed effects absorb time-varying bank-level credit supply.
- **Bank risk-taking:** the remaining variation compares how banks with different capital ratios change lending to risky firms when the overnight rate changes.
- **Selection into granted loans:** loan-application data allow the authors to estimate approval first and then outcomes conditional on approval.

4.2 Why the Spanish setting helps

- Spain is large enough to have rich bank-firm data but small enough that ECB rates are not set primarily for Spanish loan risk.
- The euro-area setting reduces concerns that Spanish credit-risk developments directly caused the short-term policy rate.
- The credit register covers almost all business loans because the reporting threshold is low.

4.3 Why low bank capital matters

- Theoretical risk-shifting is strongest when banks have little capital at stake.
- Low-capital banks may not fully internalize downside default risk.
- If lower rates reduce safe returns or make funding easier, these banks may search for yield or shift risk by expanding risky loans.

4.4 What the paper cannot fully prove

- The estimates strongly identify compositional changes in credit supply, but they do not fully separate risk-shifting from all possible search-for-yield motives.
- The setting is Spanish bank lending during a specific pre-crisis and crisis period, so external validity

requires care.

- The paper focuses on bank-firm lending decisions, not the full systemic-risk network created by correlated bank portfolios.

5 Tables and figures

5.1 Table I: Summary statistics

TABLE I

SUMMARY STATISTICS OF ALL VARIABLES USED IN THE ESTIMATIONS ON THE SAMPLE SELECTION MODEL USING LOAN APPLICATIONS[a]

Variable Name	Mean	Minimum	Q1	Median	Q3	Maximum	Standard Deviation
Dependent Variables							
I(GRANTING OF LOAN APPLICATIONS$_{tbi}$)	0.36	0	0	0	1	1	0.48
COMMITTED AMOUNT OF GRANTED LOANS$_{tbi}$	287	2	30	69	180	77,704	1315
LN(CREDIT AMOUNT$_{tbi}$)	4.41	0.69	3.40	4.23	5.19	11.26	1.32
I(FUTURE DEFAULT$_{tbi}$)	0.35	0	0	0	1	1	0.48
I(WITHOUT COLLATERAL$_{tbi}$)	0.89	0	1	1	1	1	0.31
Independent Variables							
Firm Credit Risk Variables							
Firm Credit Risk Variable With 4 Years Horizon							
I(FIRM RISK$_{ti}$)	0.03	0	0	0	0	1	0.17
Firm Credit Risk Variables With Other Horizons							
I(FIRM RISK 5 YEARS$_{ti}$)	0.03	0	0	0	0	1	0.17
I(FIRM RISK 3 YEARS$_{ti}$)	0.03	0	0	0	0	1	0.16
I(FIRM RISK 2 YEARS$_{ti}$)	0.02	0	0	0	0	1	0.14
I(FIRM RISK 1 YEAR$_{ti}$)	0.02	0	0	0	0	1	0.12
Macro Variables							
ΔOVERNIGHT RATE$_{t-1}$	0.26	−1.52	−0.08	0.23	0.74	1.48	0.71
ΔGDP$_{t-1}$	3.19	−0.27	3.00	3.46	3.83	3.98	0.88
ΔCPI$_{t-1}$	3.41	2.14	2.67	3.47	4.03	5.27	0.80
Bank Variables							
Bank Risk-Taking Variable							
BANK CAPITAL$_{t-1b}$	5.51	3.18	4.20	4.94	6.09	63.10	1.96
LN(BANK CAPITAL$_{t-1b}$)	1.66	1.16	1.44	1.60	1.81	4.14	0.30
Bank Controls							
LN(TOTAL ASSET$_{t-1b}$)	17.33	9.94	16.34	17.37	18.50	19.91	1.45
LIQUIDITY RATIO$_{t-1b}$	14.96	0.03	9.59	13.94	18.87	91.21	7.52
ROA$_{t-1b}$	0.98	−8.93	0.68	0.92	1.17	10.84	0.53
NPL RATIO$_{t-1b}$	0.90	0.00	0.36	0.62	1.03	16.08	0.89
I(COMMERCIAL BANK$_b$)	0.39	0	0	0	1	1	0.49
I(SAVINGS BANK$_b$)	0.54	0	0	1	1	1	0.50

[a]This table reports summary statistics for the sample with 241,052 observations from the 2002:02–2008:12 period. To improve the readability of the estimated coefficients, we divide the ΔOVERNIGHT RATE by 100 in all subsequent estimations.

Figure 1: Table I directly cropped from the paper.

Read.

- The first-stage sample has 241,052 loan applications from 2002:02–2008:12.
- The mean granting probability is 0.36, so roughly one-third of applications are approved.
- The mean committed amount for granted loans is 287 thousand euros; the median is 69 thousand euros.
- The future-default indicator has mean 0.35, and 0.89 of granted loans are without collateral.
- The four-year firm-risk indicator has mean 0.03, so recent nonperformers are rare but observable in a large sample.
- The annual change in the overnight rate has mean 0.26 percentage points and standard deviation 0.71.

Economic message. The sample has enough variation in approval, credit quantity, firm risk, collateral, bank capital, and rates to estimate a bank-by-firm risk-taking channel.

5.2 Table II: Main results

TABLE II
MAIN RESULTS[a]

	(1)	(2)	(3)	(4)	(5)	(6)
					BENCHMARK	TIME×BANK FE
FIRST STEP						
Dependent Variable: I(GRANTING OF LOAN APPLICATIONS$_{tbi}$)						
I(FIRM RISK$_{it}$)	−0.04***	−0.04***	−0.04***	−0.06***		
	(0.01)	(0.01)	(0.01)	(0.02)		
ΔOVERNIGHT RATE$_{t-1}$ ∗ I(FIRM RISK$_{it}$)		−2.65**	−2.62*	1.26		
	(1.22)	(1.53)	(2.37)			
ΔOVERNIGHT RATE$_{t-1}$ ∗ I(FIRM RISK$_{it}$) ∗ LN(BANK CAPITAL$_{t-1b}$)				6.97*	9.89**	9.73*
				(4.12)	(4.72)	(5.59)
Impact of 1 pp Decrease in Overnight Rate on Granting to Risky Firms						
– By the Mean Bank		*7.4%*	*7.4%*	*−3.5%*		
– By Lowly versus Highly Capitalized Bank (1 St. Dev. Difference)				*5.9%*	*8.4%*	*8.2%*
SECOND STEP						
Dependent Variable: LN(CREDIT AMOUNT$_{tbi}$)						
I(FIRM RISK$_{it}$)	−0.23***	−0.17**	−0.15	−0.30		
	(0.08)	(0.08)	(0.10)	(0.24)		
ΔOVERNIGHT RATE$_{t-1}$ ∗ I(FIRM RISK$_{it}$)		−27.39***	−25.68*	−62.59*		
		(9.93)	(13.14)	(37.51)		
ΔOVERNIGHT RATE$_{t-1}$ ∗ I(FIRM RISK$_{it}$) ∗ LN(BANK CAPITAL$_{t-1b}$)				59.56**	58.94**	53.49+
				(25.41)	(24.96)	(32.62)
Impact of 1 pp Decrease in Overnight Rate on Credit to Risky Firms						
– By the Mean Bank		*27.4%*	*25.7%*	*62.6%*		
– By Lowly versus Highly Capitalized Bank (1 St. Dev. Difference)				*18.0%*	*17.8%*	*16.1%*
Total Impact of 1 pp Decrease in Overnight Rate on Credit to Risky Firms						
– By the Mean Bank		*19.8%*	*19.2%*	*18.0%*		
– By Lowly versus Highly Capitalized Bank (1 St. Dev. Difference)				*14.6%*	*18.5%*	*17.8%*

(*Continues*)

TABLE II—*Continued*

	(1)	(2)	(3)	(4)	(5)	(6)
					BENCHMARK	TIME×BANK FE
(Year–Month) Fixed Effects	Yes	Yes	Yes	Yes	—	—
Firm Fixed Effects	No	No	No	Yes	—	—
Bank Fixed Effects	No	No	Yes	Yes	Yes	—
[(Year–Month)×Firm] Fixed Effects	No	No	No	No	Yes	Yes
[(Year–Month)×Bank] Fixed Effects	No	No	No	No	No	Yes
Bank Controls	Yes	Yes	Yes	Yes	Yes	—
{ΔGDP$_{t-1}$, ΔCPI$_{t-1}$} ∗ I(FIRM RISK$_{it}$) ∗ LN(BANK CAPITAL$_{t-1b}$)	No	No	No	Yes	Yes	Yes
ΔOVERNIGHT RATE$_{t-1}$ ∗ I(FIRM RISK$_{it}$) ∗ {Bank Controls$_{t-1b}$}	No	No	No	Yes	Yes	Yes

[a] This table reports estimates from type-2 Tobit sample selection models, which explain the probability that a loan application is approved by a bank and the loan is granted to a firm that is new to the bank (extensive margin of new lending) and the committed amount of granted loans by the bank to a firm given its loan application was successful (intensive margin of new lending). The estimates of the first step in this table come from linear probability models using ordinary least squares and 241,052 observations from the 2002:02–2008:12 period. The estimates of the second step come from the second stage of a two-step estimation procedure for panel data sample selection models outlined by Kyriazidou (1997) using kernel least squares. It uses 38,334 observations. The dependent variables are I(GRANTING OF LOAN APPLICATIONS$_{tbi}$), which equals 1 if the loan application made in month t to bank b by firm i is successful and the loan is granted in t to $t+3$, and equals 0 otherwise; and LN(CREDIT AMOUNT$_{tbi}$), which following a successful application filed in month t to bank b by firm i is the logarithm of the committed loan amount granted by bank b to firm i in t to $t+3$. The definition of the independent variables can be found in the Appendix (FIRM RISK is based on a 4-year credit history). Where possible, a constant is included but its coefficient is left unreported. When firm fixed effects are not included, firm observables are included; see the main text. Where possible, all macro, bank, and firm variables in triple interactions are included in levels and in double interactions but their coefficients are left unreported. Fixed effects are either included ("Yes"), not included ("No"), or spanned by another set of effects ("—"). For each variable the first row lists the coefficient, the second row lists the robust standard error that is corrected for multiclustering at the year–month, bank and firm level; the corresponding significance levels are adjacent to the coefficient in the second column. The single-stage/total impact is calculated as the percent change in the mean granting probability or/times the credit amount following a one percentage point decrease in the overnight interest rate and lending to firms with doubtful loans in the previous four years by a mean bank or by a lowly versus highly capitalized bank that differ by one standard deviation in capitalization. [a] The coefficient has a p-value that equals 10.1 percent. $^{+}p < 0.10$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Figure 2: Table II.

Read.

- The triple interaction is positive in both stages.
- In the benchmark model (column 5), the first-stage triple coefficient is 9.89 with two stars, and the second-stage triple coefficient is 58.94 with two stars.
- In the saturated time-by-bank fixed-effect model (column 6), the first-stage coefficient remains positive at 9.73 and the second-stage coefficient remains positive at 53.49.
- A 1 percentage point decrease in the overnight rate raises approval of risky firms by low-capital versus high-capital banks by 8.4% in the benchmark model.
- The same decrease raises committed credit to risky firms by 17.8%, and the total impact is 18.5%.

Economic message. Lower short-term rates do not merely increase total lending. They change who receives credit: low-capital banks expand risky lending more than high-capital banks.

5.3 Figures 1 and 2: Visual preview of the channel

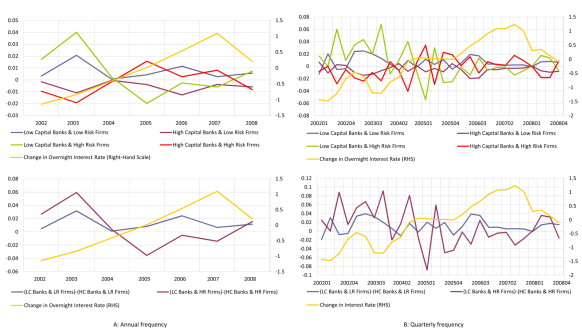

FIGURE 1.—The probability a loan application is granted (left-hand scale) by a bank with low versus high capital (by median) to an otherwise average firm with non-performing loans outstanding in the previous 4 years (HR) versus a firm without (LR), and the change in the overnight interest rate (right-hand scale).

FIGURE 2.—The logarithm of the committed amount of granted loans (left-hand scale) by a bank with low versus high capital (by median) to an otherwise average firm with non-performing loans outstanding in the previous 4 years (HR) versus a firm without (LR), and the change in the overnight interest rate (right-hand scale).

Figure 3: Figures 1 and 2 grouped side-by-side.

Read.

- Figure 1 plots approval probabilities for low- versus high-capital banks and high- versus low-risk firms, together with the change in the overnight rate.
- Figure 2 repeats the exercise for the log committed amount of granted loans.

- The plots show that when the overnight rate falls, the low-capital-bank/high-risk-firm series moves more strongly than the high-capital-bank/high-risk-firm series.

Economic message. The figures visually preview the main regression result: falling overnight rates are associated with more risky credit supply by low-capital banks.

5.4 Table III: Various robustness and macro horse races

TABLE III

VARIOUS ROBUSTNESS[a]

	(1)	(2)	(3)	(4)	(5)	(6)
	BENCHMARK+ ΔCPI$_{t-1}$ IN ALL INT'ACTIONS	BENCHMARK+ ΔSPANISH LONG-TERM INTEREST RATE	BENCHMARK+ ΔU.S. SHORT (LONG) TERM INTEREST RATE	BENCHMARK+ ΔSPANISH SECURITIZED ASSETS/TOT. ASSETS	TIME+BANK FE+ CTRLS OF (4) & A.III (5) & (6)	TIME+BANK FE NO CORRECTION FOR SAMPLE SELECTION: ONE-STEP
FIRST STEP						
Dependent Variable: I(GRANTING OF LOAN APPLICATIONS$_{ibt}$)						
ΔOVERNIGHT RATE$_{t-1}$ $*$ I(FIRM RISK$_{it}$) $*$ LN(BANK CAPITAL$_{t-1b}$)	9.34**		10.86**	10.05**	10.82*	
	(4.68)		(4.58)	(4.64)	(5.75)	
CONTROL$_{t-1}$ $*$ I(FIRM RISK$_{it}$) $*$ LN(BANK CAPITAL$_{t-1b}$)		3.41	4.67*	0.85		
		(6.04)	(2.73)	(5.95)		
For Lowly versus Highly Capitalized Bank (1 St. Dev. Difference)						
– Impact of 1 pp Decrease in Overnight Rate on Granting to Risky Firms	*7.9%*		*9.2%*	*8.5%*	*9.2%*	
– Impact of 1 pp Decrease in Control on Granting to Risky Firms		*2.9%*	*4.0%*	*0.7%*		
SECOND STEP						
Dependent Variable: LN(CREDIT AMOUNT$_{ibt}$)						
ΔOVERNIGHT RATE$_{t-1}$ $*$ I(FIRM RISK$_{it}$) $*$ LN(BANK CAPITAL$_{t-1b}$)	82.88***		86.96***	55.62*	86.88**	31.39
	(23.18)		(30.27)	(30.29)	(41.74)	(28.11)
CONTROL$_{t-1}$ $*$ I(FIRM RISK$_{it}$) $*$ LN(BANK CAPITAL$_{t-1b}$)		10.08	49.10**	−15.26		
		(29.37)	(24.11)	(42.78)		
For Lowly versus Highly Capitalized Bank (1 St. Dev. Difference)						
– Impact of 1 pp Decrease in Overnight Rate on Credit to Risky Firms	*25.0%*		*26.2%*	*16.8%*	*26.2%*	*9.5%*
– Impact of 1 pp Decrease in Control on Credit to Risky Firms		*3.0%*	*14.8%*	*−4.6%*		
For Lowly versus Highly Capitalized Bank (1 St. Dev. Difference)						
– Total Impact of 1 pp Decrease in Overnight Rate on Credit to Risky Firms	*19.8%*		*22.3%*	*18.5%*	*22.2%*	*13.2%*
– Total Impact of 1 pp Decrease in Control on Credit to Risky Firms		*5.5%*	*10.6%*	*−0.2%*		

(*Continues*)

TABLE III—*Continued*

	(1)	(2)	(3)	(4)	(5)	(6)
	BENCHMARK+ ΔCPI$_{t-1}$ IN ALL INT'ACTIONS	BENCHMARK+ ΔSPANISH LONG-TERM INTEREST RATE	BENCHMARK+ ΔU.S. SHORT (LONG) TERM INTEREST RATE	BENCHMARK+ ΔSPANISH SECURITIZED ASSETS/TOT. ASSETS	TIME+BANK FE+ CTRLS OF (4) & A.III (5) & (6)	TIME+BANK FE NO CORRECTION FOR SAMPLE SELECTION: ONE-STEP
[(Year–Month)+Firm] Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Bank Fixed Effects	Yes	Yes	Yes	Yes	—	—
[(Year–Month)+Bank] Fixed Effects	No	No	No	No	Yes	Yes
Bank Controls	Yes	Yes	Yes	Yes	Yes	—
{ΔGDP$_{t-1}$, ΔCPI$_{t-1}$} $*$ I(FIRM RISK$_{it}$) $*$ LN(BANK CAPITAL$_{t-1b}$)	Yes	Yes	Yes	Yes	Yes	Yes
ΔOVERNIGHT RATE$_{t-1}$ $*$ I(FIRM RISK$_{it}$) $*$ {Bank Controls$_{t-1b}$}	Yes	Yes	Yes	Yes	Yes	Yes

[a] This table reports in models (1)–(5) the estimates from type-2 Tobit sample selection models, which explain the probability that a loan application is approved by a bank and the loan is granted to a firm that is new to the bank (extensive margin of new lending) and the committed amount of granted loans by the bank to a firm given its loan application was successful (intensive margin of new lending). The estimates of the first step in this table come from linear probability models using ordinary least squares and 241,052 observations from the 2002:02–2008:12 period. The estimates of the second step come from the second stage of a two-step estimation procedure for panel data sample selection models outlined by Kyriazidou (1997) using kernel least squares. It uses 38,334 observations. The estimates in model (6) come from a one-step ordinary least squares regression. The dependent variables are I(GRANTING OF LOAN APPLICATIONS$_{ibt}$), which equals 1 if the loan application made in month t to bank b by firm i is successful and the loan is granted in t to $t+3$, and equals 0 otherwise; and LN(CREDIT AMOUNT$_{ibt}$), which following a successful application filed in month t to bank b by firm i is the logarithm of the committed loan amount granted in t to $t+3$ by bank b to firm i. The definition of the independent variables can be found in the Appendix (FIRM RISK is based on a 4-year credit history). Where possible, a constant is included but its coefficient is left unreported. Where possible, all macro, bank, and firm variables in triple interactions are included in levels and in double interactions but their coefficients are left unreported. Fixed effects are either included ("Yes"), not included ("No"), or spanned by another set of effects ("—"). For each variable, the first row lists the coefficient, the second row lists the robust standard error that is corrected for multiclustering at the year-month, bank, and firm level; the corresponding significance levels are adjacent to the coefficient in the second column. The single-stage/total impact is calculated as the percent change in the mean granting probability or times the credit amount following a 1 percentage point decrease in the overnight interest rate or control variable and lending to firms with doubtful loans in the previous four years by a lowly versus highly capitalized bank that differ by 1 standard deviation in capitalization. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Figure 4: Table III.

Read.

- The overnight-rate triple remains positive and significant after adding CPI interactions, Spanish long-term rates, U.S. rates, securitization, current-account controls, and time-by-bank fixed effects.
- In column 5, the first-stage triple coefficient is 10.82 and the second-stage coefficient is 86.88.
- In column 5, a 1 percentage point decrease in the overnight rate implies a 9.2% higher approval effect and 26.2% higher credit-amount effect for low- versus high-capital banks lending to risky firms.
- The one-step model without correcting for sample selection produces a smaller total effect, illustrating downward bias when the approval stage is ignored.

Economic message. The result is not explained by long-term rates, securitization, U.S. monetary conditions, or capital inflows. The application stage matters for measuring the channel.

5.5 Table IV: Robustness to the firm-risk horizon

TABLE IV

ROBUSTNESS WITH RESPECT TO THE TIME HORIZON ON NONPERFORMING LOANS TO MEASURE FIRM CREDIT RISK[a]

	(1)	(2)	(3)	(4)	(5)
FIRM CREDIT RISK Time Horizon:	5 YEARS	BENCHMARK: 4 YEARS	3 YEARS	2 YEARS	1 YEAR
FIRST STEP					
Dependent Variable: I(GRANTING OF LOAN APPLICATIONS$_{ibt}$)					
ΔOVERNIGHT RATE$_{t-1}$ $*$ I(FIRM RISK$_{it}$) $*$ LN(BANK CAPITAL$_{t-1b}$)	9.05*	9.89**	10.93**	13.49***	8.49
	(4.87)	(4.72)	(5.27)	(5.22)	(6.43)
Impact of 1 pp Decrease in Overnight Rate on Granting to Risky Firms by Lowly versus Highly Capitalized Bank (1 St. Dev. Difference)	*7.7%*	*8.4%*	*9.2%*	*11.4%*	*7.2%*
SECOND STEP					
Dependent Variable: LN(CREDIT AMOUNT$_{ibt}$)					
ΔOVERNIGHT RATE$_{t-1}$ $*$ I(FIRM RISK$_{it}$) $*$ LN(BANK CAPITAL$_{t-1b}$)	58.31***	58.94**	45.34**	42.95*	88.02***
	(17.91)	(24.96)	(22.67)	(22.38)	(31.38)
Impact of 1 pp Decrease in Overnight Rate on Credit to Risky Firms by Lowly versus Highly Capitalized Bank (1 St. Dev. Difference)	*17.6%*	*17.8%*	*13.7%*	*12.9%*	*26.5%*
Total Impact of 1 pp Decrease in Overnight Rate on Credit to Risky Firms by Lowly versus Highly Capitalized Bank (1 St. Dev. Difference)	*17.3%*	*18.5%*	*18.8%*	*22.0%*	*7.6%*

(*Continues*)

TABLE IV—*Continued*

	(1)	(2)	(3)	(4)	(5)
FIRM CREDIT RISK Time Horizon:	5 YEARS	BENCHMARK: 4 YEARS	3 YEARS	2 YEARS	1 YEAR
[(Year–Month)+Firm] Fixed Effects	Yes	Yes	Yes	Yes	Yes
Bank Fixed Effects	Yes	Yes	Yes	Yes	Yes
Bank Controls	Yes	Yes	Yes	Yes	Yes
{ΔGDP$_{t-1}$, ΔCPI$_{t-1}$} $*$ I(FIRM RISK$_{it}$) $*$ LN(BANK CAPITAL$_{t-1b}$)	Yes	Yes	Yes	Yes	Yes
ΔOVERNIGHT RATE$_{t-1}$ $*$ I(FIRM RISK$_{it}$) $*$ {Bank Controls$_{t-1b}$}	Yes	Yes	Yes	Yes	Yes

[a] This table reports estimates from type-2 Tobit sample selection models, which explain the probability that a loan application is approved by a bank and the loan is granted to a firm that is new to the bank (extensive margin of new lending) and the committed amount of granted loans by the bank to a firm given its loan application was successful (intensive margin of new lending). The estimates of the first step in this table come from linear probability models using ordinary least squares and 241,052 observations from the 2002:02–2008:12 period. The estimates of the second step come from the second stage of a two-step estimation procedure for panel data sample selection models outlined by Kyriazidou (1997) using kernel least squares. It uses 38,334 observations. The dependent variables are I(GRANTING OF LOAN APPLICATIONS$_{ibt}$), which equals 1 if the loan application made in month t to bank b by firm i is successful and the loan is granted in t to $t+3$, and equals 0 otherwise; and LN(CREDIT AMOUNT$_{ibt}$), which following a successful application filed in month t to bank b by firm i is the logarithm of the committed loan amount granted in t to $t+3$ by bank b to firm i. The definition of the independent variables can be found in the Appendix (FIRM RISK is based on a 4-year credit history). Where possible, a constant is included but its coefficient is left unreported. Where possible, all macro, bank, and firm variables in triple interactions are included in levels and in double interactions but their coefficients are left unreported. Fixed effects are either included ("Yes"), not included ("No"), or spanned by another set of effects ("—"). For each variable the first row lists the coefficient, the second row lists the robust standard error that is corrected for multiclustering at the year-month, bank, and firm level; the corresponding significance levels are adjacent to the coefficient in the second column. The single-stage/total impact is calculated as the percent change in the mean granting probability or times the credit amount following a 1 percentage point decrease in the overnight interest rate and lending to firms with doubtful loans in the previous 4 years by a lowly versus highly capitalized bank that differ by 1 standard deviation in capitalization. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Figure 5: Table IV.

Read.

- The authors redefine firm risk using nonperforming-loan histories from one to five years.
- The approval impact ranges from 7.2% to 11.4%.
- The committed-credit impact ranges from 12.9% to 26.5%.
- The total impact remains positive across the alternative risk windows.

Economic message. The main result is not an artifact of choosing a four-year window to measure past nonperformance.

5.6 Tables V and VI: Future default and collateral

TABLE V

THE PROBABILITY THAT A FIRM BECOMES DELINQUENT WITH THE BANK IN THE FUTURE[a]

	(1)	(2)	(3)	(4)	(5)	(6) ONE-STEP
FIRST STEP						
Dependent Variable: I(GRANTING OF LOAN APPLICATIONS$_{ibt}$)						
ΔOVERNIGHT RATE$_{t-1}$	−2.04	−6.31				
	(4.58)	(4.59)				
LN(BANK CAPITAL$_{t-1b}$)	−0.02	0.05	0.05	0.06	0.28***	
	(0.04)	(0.05)	(0.05)	(0.06)	(0.08)	
ΔOVERNIGHT RATE$_{t-1}$ ∗ LN(BANK CAPITAL$_{t-1b}$)	2.09*	0.81	0.84	0.58	0.10	
	(1.22)	(1.37)	(1.34)	(1.54)	(1.51)	
SECOND STEP						
Dependent Variable: I(FUTURE DEFAULT$_{ibt}$)						
ΔOVERNIGHT RATE$_{t-1}$	−16.37*	−37.63***				
	(9.93)	(8.89)				
LN(BANK CAPITAL$_{t-1b}$)	−0.11	−0.02	−0.02	−0.05	−0.14	2.70
	(0.08)	(0.11)	(0.11)	(0.12)	(0.19)	(1.70)
ΔOVERNIGHT RATE$_{t-1}$ ∗ LN(BANK CAPITAL$_{t-1b}$)	3.77*	3.93**	3.59*	6.17***	6.27***	3.20*
	(2.26)	(1.99)	(2.12)	(2.39)	(2.42)	(1.84)
Impact of 1 pp Decrease in Overnight Rate on Future Default by Lowly versus Highly Capitalized Bank (1 St. Dev. Difference)	*3.2%*	*3.3%*	*3.1%*	*5.3%*	*5.3%*	*2.7%*
Total Impact of 1 pp Decrease in Overnight Rate on Future Default by Lowly versus Highly Capitalized Bank (1 St. Dev. Difference)	*2.6%*	*1.0%*	*0.9%*	*1.5%*	*1.5%*	*0.8%*

(*Continues*)

TABLE V—*Continued*

	(1)	(2)	(3)	(4)	(5)	(6) ONE-STEP
(Year–Month) Fixed Effects	No	No	Yes	—	—	—
Firm Fixed Effects	Yes	Yes	Yes	—	—	—
Bank Fixed Effects	No	Yes	Yes	Yes	Yes	Yes
[(Year–Month)∗Firm] Fixed Effects	No	No	No	Yes	Yes	Yes
PERIOD IN WHICH FIRM CAN DEFAULT and PERIOD SQUARED	Yes	Yes	—	—	—	—
Bank Controls	Yes	Yes	Yes	Yes	Yes	Yes
{ΔGDP$_{t-1}$, ΔCPI$_{t-1}$} ∗ LN(BANK CAPITAL$_{t-1b}$)	Yes	Yes	Yes	Yes	Yes	Yes
{ΔU.S. Short-Term IR$_{t-1}$, ΔSecurit./TA$_{t-1}$, ΔCurr. Acc./GDP$_{t-1}$} ∗ LN(BANK CAPITAL$_{t-1b}$)	No	No	No	No	Yes	No
ΔOVERNIGHT RATE$_{t-1}$ ∗ {Bank Controls}	Yes	Yes	Yes	Yes	Yes	Yes

[a]This table reports in models (1)–(5) the estimates from type-2 Tobit sample selection models, which explain the probability that a loan application is approved by a bank and the loan is granted to a firm that is new to the bank (extensive margin of new lending) and subsequently the firm defaults (risk-taking margin of new lending). The estimates of the first step in this table come from linear probability models using ordinary least squares and 241,052 observations from the 2002:02–2008:12 period. The estimates of the second step come from the second stage of a two-step estimation procedure for panel data sample selection models outlined by Kyriazidou (1997) using kernel least squares. It uses 38,334 observations. The estimates in model (6) come from a one-step ordinary least squares regression. The dependent variables are I(GRANTING OF LOAN APPLICATIONS$_{ibt}$), which equals 1 if the loan application made in month t to bank b by firm i is successful and the loan is granted in t to $t+3$, and equals 0 otherwise; and I(FUTURE DEFAULT$_{ibt}$), which equals 1 when firm i that is granted the loan at time t by bank b defaults at some point in the future. The definition of the independent variables can be found in the Appendix (FIRM RISK is based on a 4-year credit history). Where possible, a constant is included but its coefficient is left unreported. Where possible, all macro and bank level variables in double interactions and the double interactions reported in the table are included in levels but their coefficients are left unreported. Fixed effects are either included ("Yes"), not included ("No"), or spanned by another set of effects ("—"). For each variable the first row lists the coefficient, the second row lists the robust standard error that is corrected for multiclustering at the year–month, bank, and firm level; the corresponding significance levels are adjacent to the coefficient in the second column. The single-stage (total) impact is calculated as the percent change in the mean default probability (times the probability a loan application is granted) following on a 1 percentage point decrease in the overnight interest rate for a lowly versus highly capitalized bank that differ by 1 standard deviation in capitalization. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Figure 6: Table V.

TABLE VI

LOANS GRANTED WITHOUT COLLATERAL[a]

	(1) Table II Model (5)	(2) Table III Model (5)	(3) Table III Model (6) ONE-STEP
FIRST STEP			
Dependent Variable: I(GRANTING OF LOAN APPLICATIONS$_{ibt}$)			
ΔOVERNIGHT RATE$_{t-1}$ ∗ I(FIRM RISK$_{it}$) ∗ LN(BANK CAPITAL$_{t-1b}$)	9.89**	10.82*	
	(4.72)	(5.75)	
Impact of 1 pp Decrease in Overnight Rate on Uncollateralized Credit to Risky Firms by Lowly versus Highly Capitalized Bank (1 St. Dev. Difference)	*8.4%*	*9.2%*	
SECOND STEP			
Dependent Variable: I(WITHOUT COLLATERAL$_{ibt}$)			
ΔOVERNIGHT RATE$_{t-1}$ ∗ I(FIRM RISK$_{it}$) ∗ LN(BANK CAPITAL$_{t-1b}$)	7.94	19.60*	12.25
	(9.80)	(11.69)	(9.77)
Impact of 1 pp Decrease in Overnight Rate on Uncollateralized Credit to Risky Firms by Lowly versus Highly Capitalized Bank (1 St. Dev. Difference)	*2.7%*	*6.6%*	*4.1%*
Total Impact of 1 pp Decrease in Overnight Rate on Uncollateralized Credit to Risky Firms by Lowly versus Highly Capitalized Bank (1 St. Dev. Difference)	*3.5%*	*4.8%*	*4.1%*

(*Continues*)

TABLE VI—*Continued*

	(1) Table II Model (5)	(2) Table III Model (5)	(3) Table III Model (6) ONE-STEP
Bank Fixed Effects	Yes	—	—
[(Year–Month)∗Firm] Fixed Effects	Yes	Yes	Yes
[(Year–Month)∗Bank] Fixed Effects	No	Yes	Yes
ΔOVERNIGHT RATE$_{t-1}$ ∗ I (FIRM RISK$_{it}$) ∗ {Bank Controls$_{t-1b}$}	Yes	Yes	Yes

[a]This table reports in models (1) and (2) the estimates from type-2 Tobit sample selection models, which explain the probability that a loan application is approved by a bank and the loan is granted to a firm that is new to the bank (extensive margin of new lending) and the lack of collateralization of the loan granted by the bank to a firm given its loan application was successful (risk-taking margin of new lending). The estimates of the first step in this table come from linear probability models using ordinary least squares and 241,052 observations from the 2002:02–2008:12 period. The estimates of the second step come from the second stage of a two-step estimation procedure for panel data sample selection models outlined by Kyriazidou (1997) using kernel least squares. It uses 38,334 observations. The estimates in model (3) come from a one-step ordinary least squares regression. The dependent variables are I(GRANTING OF LOAN APPLICATIONS$_{ibt}$), which equals 1 if the loan application made in month t to bank b by firm i is successful and the loan is granted in t to $t+3$, and equals 0 otherwise; and I(WITHOUT COLLATERAL$_{ibt}$), which equals 1 if the loan granted in month t by bank b by firm i is uncollateralized, and equals 0 otherwise. The definition of the independent variables can be found in the Appendix (FIRM RISK is based on a 4-year credit history). Where possible, a constant is included but its coefficient is left unreported. Where possible, all macro and bank level variables in double interactions and the double interactions reported in the table are included in levels but their coefficients are left unreported. Fixed effects are either included ("Yes"), not included ("No"), or spanned by another set of effects ("—"). For each variable the first row lists the coefficient, the second row lists the robust standard error that is corrected for multiclustering at the year–month, bank and firm level; the corresponding significance levels are adjacent to the coefficient in the second column. The single-stage/total impact is calculated as the percent change in the mean granting probability or/times the probability of uncollateralized credit following a one percentage point decrease in the overnight interest rate for a lowly versus highly capitalized bank (that differ by one standard deviation in capitalization). * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Figure 7: Table VI.

Read.

- Table V shows that when the overnight rate falls, low-capital banks' granted loans have a higher future default probability.
- Depending on specification, the effect for low- versus high-capital banks is about 3.1%–5.3% at the mean.
- Table VI shows that the same channel is associated with more loans without collateral.
- In the richer Table VI specification, the second-stage uncollateralized-credit effect is 6.6%, and the total effect is 4.8%.

Economic message. The paper does not only find more lending to ex ante risky firms. It also finds worse ex post loan quality and weaker collateral, consistent with excessive risk-taking.

5.7 Table VII: Variable definitions

APPENDIX

TABLE VII

DEFINITIONS OF ALL VARIABLES USED IN THE ESTIMATIONS

Variable Name	Unit	Definition
Dependent Variables		
I(GRANTING OF LOAN APPLICATIONS$_{ibt}$)	0/1	A dummy variable, which equals 1 if the loan application made in month t to bank b by firm i is successful and the loan is granted in t to $t+3$, and equals 0 otherwise
LN(CREDIT AMOUNT$_{ibt}$)	ln (000 Euros)	The logarithm of the committed loan amount granted in months t to $t+3$ by bank b to firm i following a successful application filed in month t to bank b by firm i
I(FUTURE DEFAULT$_{ibt}$)	0/1	A dummy variable, which equals 1 when firm i that is granted the loan in month t by bank b defaults at some point in the future, and equals 0 otherwise
I(WITHOUT COLLATERAL$_{ibt}$)	0/1	A dummy variable, which equals 1 if the loan granted in month t by bank b by firm i is uncollateralized, and equals 0 otherwise
Independent Variables		
Firm-Level Credit Risk Variables		
I(FIRM RISK$_{it}$)	0/1	A dummy variable, which equals 1 if in month t the firm had nonperforming loans outstanding 4 years prior to the loan application, and equals 0 otherwise
Macro-Level Variables		
ΔOVERNIGHT RATE$_{t-1}$	%	The annual change in the Euro overnight index average rate (EONIA), which is the target interest rate for monetary policy in the Eurosystem, and before 1998 the change in the Spanish overnight interest rate at $t-1$
ΔGDP$_{t-1}$	%	Annual change of Spanish gross domestic product in real terms at $t-1$
ΔCPI$_{t-1}$	%	Annual change of Spanish Consumer Price Index at $t-1$

(*Continues*)

TABLE VII—*Continued*

Variable Name	Unit	Definition
Additional Macro-Level Controls		
ΔSPANISH LONG-TERM INTEREST RATE$_{t-1}$	%	The annual change in the 10-year Spanish government bond rate at $t-1$
ΔU.S. SHORT-TERM INTEREST RATE$_{t-1}$	%	The annual change in the U.S. Federal funds rate at $t-1$
ΔU.S. LONG-TERM INTEREST RATE$_{t-1}$	%	The annual change in the 10-year U.S. government bond rate at $t-1$
ΔSPANISH SECURITIZED ASSETS/TOT. ASSETS$_{t-1}$	%	The annual change in assets that were securitized bay Spanish banks at $t-1$
ΔSPANISH CURRENT ACCOUNT DEFICIT/GDP$_{t-1}$	%	The annual change in the Spanish current account deficit over GDP at $t-1$
Bank-Level Variables		
Bank Risk-Taking Variable		
LN(BANK CAPITAL$_{t-1b}$)	—	The logarithm of the ratio of bank equity over total assets of the bank at $t-1$
Bank Controls		
LN(TOTAL ASSETS$_{t-1b}$)	ln (000 Euros)	The logarithm of the total assets of the bank
LIQUIDITY RATIO$_{t-1b}$	%	The ratio of liquid assets (cash and balance with central banks, and loans and advances to governments and credit institutions) held by the bank over the total assets of the bank
ROA$_{t-1b}$	%	The total net income over assets of the bank
NPL RATIO$_{t-1b}$	%	The nonperforming loan ratio of the bank
I(COMMERCIAL BANK$_b$)	0/1	A dummy variable, which equals 1 if the bank is a commercial bank and equals 0 otherwise
I(SAVINGS BANK$_b$)	0/1	A dummy variable, which equals 1 if the bank is a savings bank and equals 0 otherwise

Figure 8: Appendix Table VII.

Read.

- Table VII defines the dependent variables, firm-risk variables, macro controls, and bank-level variables.
- The table clarifies that the key credit-risk measure is an indicator for recent nonperforming loans.
- It also clarifies the construction of the overnight-rate change, Spanish long-term rate change, U.S. rate changes, securitization, current account, and bank characteristics.

Economic message. The appendix definitions are important because the paper's identification relies on precise distinctions among firm risk, bank capital, macro rates, and alternative macro channels.

6 Short summary

- The paper asks whether low monetary policy rates increase bank risk-taking.
- It uses unusually rich Spanish credit-register data on loan applications and outcomes.
- The central design compares multiple banks' decisions on the same firm in the same month and, in the most saturated model, the same bank's lending to different-risk firms in the same month.
- The core coefficient is the triple interaction of overnight-rate change, firm risk, and bank capital.
- Lower overnight rates make low-capital banks approve more applications by risky firms.
- Conditional on approval, low-capital banks commit larger amounts to risky firms.
- These loans are more likely to be uncollateralized and to default later.
- Long-term interest rates and other macro variables do not explain away the findings.
- Ignoring the application stage understates the effect because selection into granted loans is itself a form of risk-taking.
- The paper provides direct evidence that monetary policy changes the risk composition of credit supply.

7 Conclusion

7.1 Core conclusion

A lower overnight interest rate induces banks to take more credit risk, especially banks with lower capital. The effect appears in loan approval, loan volume, collateral requirements, and future default.

7.2 Economic intuition

Low-capital banks have less capital at stake and may not fully internalize downside default risk. When short-term rates fall, these banks have stronger incentives to search for yield or shift risk by expanding credit to risky firms.

7.3 Takeaway

The policy-relevant object is not only how much credit is created, but who receives it and under what terms. Monetary policy can therefore interact with financial stability through the composition of bank credit supply.